

Answer **all** questions.

Question 1 [15 marks]

The many species of orchids comprise one of the largest families of flowering plants, the Orchidaceae, which contains about 1,000 genera and about 20,000 species. With the recent advances in genetic engineering, several attempts have been made to create tri-species hybrids where desirable genes from three different species are incorporated into one plant. In one of these attempts, a horticulturist selected the following species for a particular trait that each exhibits: *Brassavola nodosa* (for vigorous growth), *Vanda coerulea* (for its fragrance) and *Dendrobium aggregatum* (for long flowering periods). In another attempt to create tri-species hybrids, he replaced the dendrobium with *Vanda javierii* (for large blooms).

In both attempts, the gene of interest was isolated from the brassavola and inserted into a plasmid to produce the recombinant DNA, which was then introduced into pollen grains of *Vanda coerulea*. As pollen grains have a tough outer coat, the recombinant DNA is introduced into pollen tubes which grow out of germinated pollen grains. Thus the pollen is put into a culture solution that causes the grains to germinate and recombinant DNA is added once the pollen tubes are produced. The mixture is placed in a tube containing two parallel electrodes and a large electric field is applied between these electrodes in 80 microsecond pulses that produce tiny holes in the sides of the pollen tubes. The holes remain open for as long as 30 minutes and DNA can enter the pollen cells through the pores. The altered pollen grains are then placed on the stigmatic surfaces of *Dendrobium aggregatum* and *Vanda javierii*.

After a period of 2 weeks, seed pods were produced on *Vanda javierii* but not on *Dendrobium aggregatum*.

(a) Explain the role of **named** enzymes in the production of recombinant DNA.

[4]

(b) Name the method used to introduce the recombinant DNA into the host's pollen grains.

[1]

- (c) State and describe two other methods which can be used to introduce the recombinant DNA into the host's pollen grains.

[4]

- (d) Suggest a way to identify the pollen grains of *Vanda coerulea* that have been transformed.

[3]

- (e) Suggest why seed pods were not produced on *Dendrobium aggregatum*.

[1]

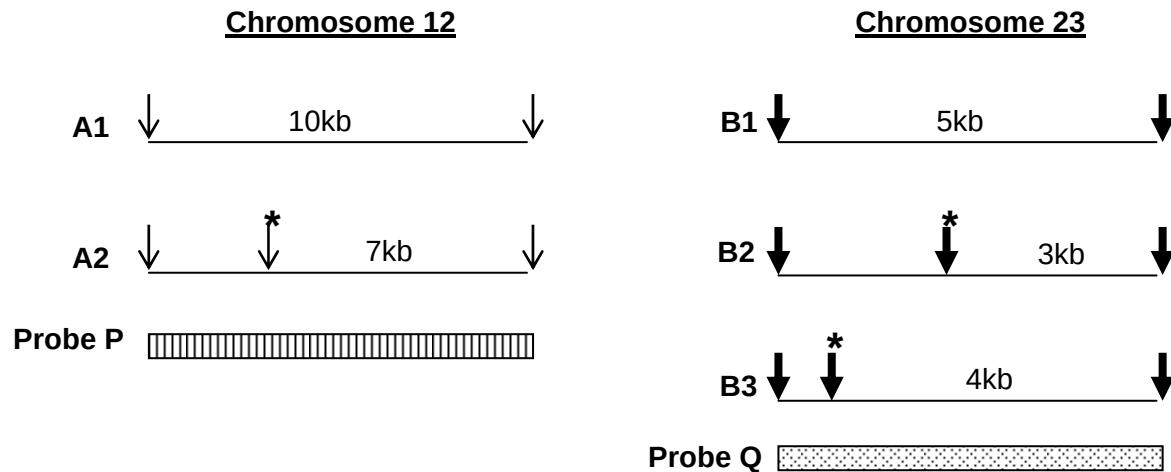
- (f) For a **named** crop plant, suggest two genetic modifications that might profitably be made.

[2]

Question 2 [13 marks]

The DNA of a family affected by a rare autosomal disease was analyzed using RFLP. Two genes suspected to be responsible for the disease were analyzed. Fig. 2.1 shows the two genes, A and B, which are found on chromosomes 12 and 23 respectively.

Eco RI was used to restrict Locus A and two types of alleles, A1 and A2 were detected. *Pst I* was used to restrict Locus B and three types of alleles B1, B2 and B3 were detected. The RFLP in locus A and B were identified by probes P and Q respectively.



Polymorphic restriction sites were found at each locus indicated with (*).

↓ indicates *Eco RI* restriction sites.

↓ indicates *Pst I* restriction sites.

Fig. 2.1

- (a) With reference to Fig. 2.1, explain how the two alleles of gene A can be distinguished using gel electrophoresis.

[3]

Fig. 2.2 shows the pedigree chart and RFLP analysis of a family affected by the disease. A phenotypically normal couple had three children W, X and Y, with only one of them affected by the disease (indicated by the shaded square).

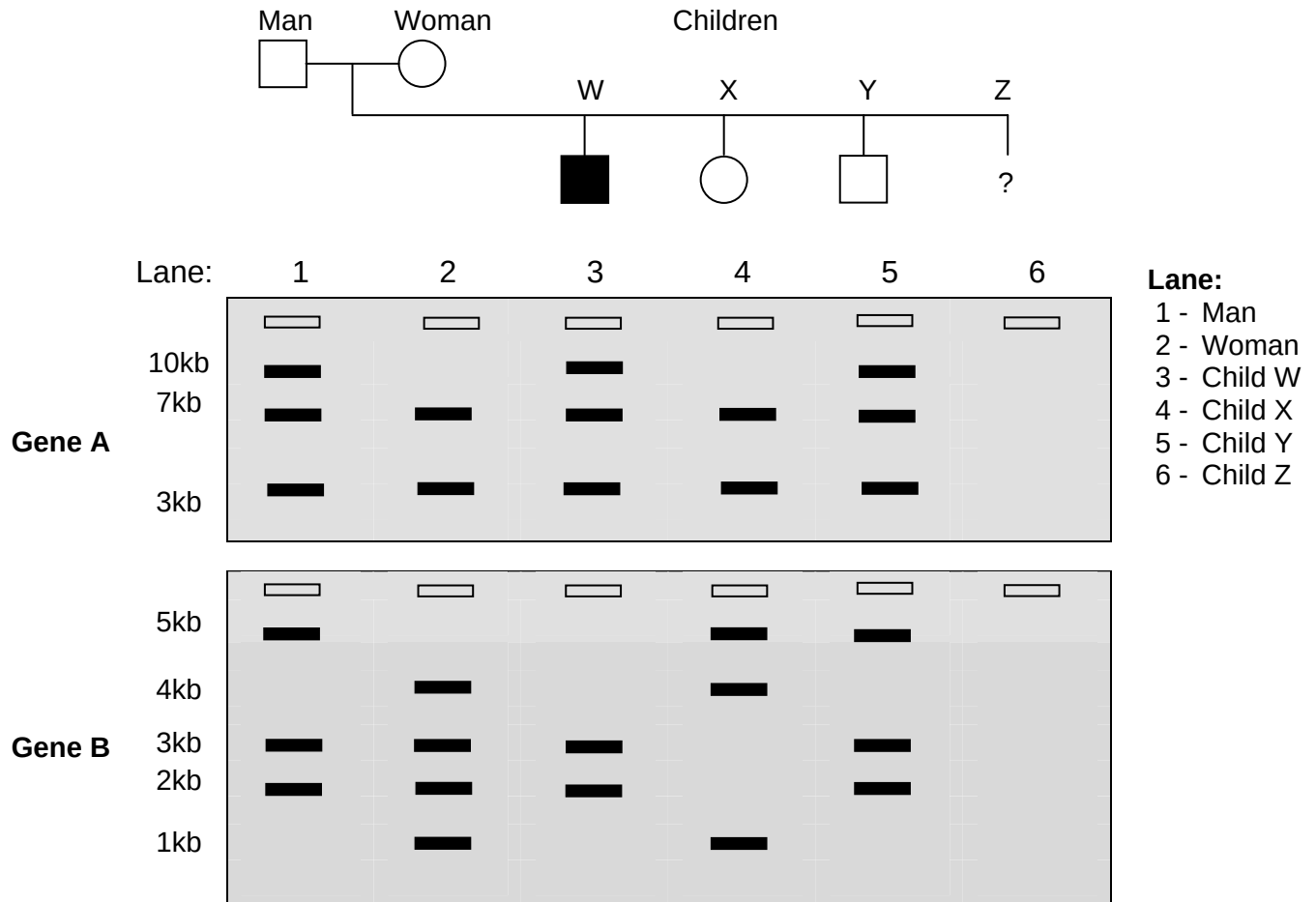


Fig. 2.2

(b) With reference to Fig. 2.2,

(i) state which gene caused the disease.

[1]

(ii) state the genotype of Child W and explain the inheritance of the disease. Use the symbols A1, A2, B1, B2 and B3 to represent the different alleles shown in Fig 2.1.

[3]

The couple recently gave birth to a fourth child, Child Z who was found to be homozygous at both gene loci.

(c) In Fig. 2.2, indicate the bands that would be obtained for Child Z in Lane 6. [1]

(d) State **two** features of gene probes that can be used to detect RFLP.

[2]

(e) With reference to RFLP,

(i) define genetic markers

[1]

(ii) explain the features of genetic markers in analyzing diseases.

[2]

Question 3 [17 marks]

All types of blood cells develop from haemopoietic (blood forming) stem cells that originate in the bone marrow. These stem cells are said to be multipotent. Damage to stem cells affects a person's ability to produce appropriate numbers of fully functioning blood cells and platelets. Transplant bone marrow can be used to treat a patient whose own marrow contains damaged stem cells. Bone marrow need not be donated by someone else. In some circumstances, stem cells from the patient's own marrow can be used for treatment. Stem cells are collected and stored prior to the patient receiving chemotherapy or radiotherapy. The stored cells are cultured and treated before being reintroduced back into the patient. Recent developments for stem cell technology include removal and storage of stem cells from the umbilical cord at birth.

(a) Describe the unique features of stem cells.

[3]

(b) State **one** main feature of a multipotent stem cell.

[1]

(c) Explain the meaning of the term *differentiation*.

[1]

(d) Stem cells collected from the patient, for future use, may include malignant (cancerous) cells. Suggest how these cells might be treated before transplanting them back into the patient.

[1]

Removal of stem cells from the umbilical cord at birth and their storage in cord blood bank appear to be a good insurance against future diseases for these individuals.

- (e) Suggest one advantage and one disadvantage of using umbilical cord stem cells for treating diseases in the same individual.

[2]

Cystic fibrosis (CF) is a genetic disease which affects many parts of the body. This disease is caused by mutations of a gene coding for a transmembrane protein (CFTR) which acts as an ion pore.

A study on ion transport was carried out on pancreatic cells by comparing ion transport in cells expressing the normal allele for CFTR with that in cells expressing mutant alleles. The mutant cells chosen for study were all capable of producing a CFTR protein and correctly inserting it into their cell surface membrane. This was observed in both types of CF with adequate and inadequate functioning of the pancreas.

The abilities of the cells to transport HCO_3^- and Cl^- were measured and expressed as a $\text{HCO}_3^- : \text{Cl}^-$ transport ratio. The results are shown in Table 3.1.

Table 3.1

CF allele expressed by cells	$\text{HCO}_3^- : \text{Cl}^-$ transport ratio of cells
Normal alleles	1.0 : 1.0
Mutant alleles associated with CF with inadequate functioning of the pancreas	All < 0.1 : 1.0
Mutant alleles associated with CF with adequate functioning of the pancreas	0.3 – 0.46 : 1.0

- (f) With reference to Table 3.1, explain the cause of inadequate functioning of the pancreas in CF.

[2]

- (g) Suggest **one** possible consequence of inadequate functioning of the pancreas in CF.

[1]

Conventional treatments for cystic fibrosis include the use of physiotherapy, antibiotics and pancreatic supplements. Recent developments include gene therapy in particular targeting lung tissue. Initial gene therapy trials involving lung epithelial cells presented positive results which were short-lived due to short lifespan of epithelial cells. Recent trials included using haemopoietic stem cells from the bone marrow or cord blood as well as the use of lung stem cells.

(h) Briefly explain the genetic basis of gene therapy.

[2]

(i) State **two** factors that may keep gene therapy from becoming an effective treatment for genetic diseases.

[2]

(j) State **two** ethical and social concerns regarding the use of gene therapy.

[2]

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 4 **(a)** Outline the procedure for cloning a **named** eukaryotic gene using bacterial transformation. [8]
- (b)** Explain the advantages of plant tissue culture. [6]
- (c)** Discuss the ethical and social implications of genetically modified organisms. [6]

